Department of Computer Science

CPSI 77503: Machine Learning

Section 01: MW 10:50 AM -- 12:05 PM in EI325

Fall 2026

Machine Learning (01)

Instructor Info

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Course Description

Credit Hour(s): 3

Description:

An introduction to the foundations and practice of machine learning. The course begins by placing machine learning within the broader field of artificial intelligence and introduces concepts from search, reasoning under uncertainty, and optimization that support modern learning algorithms. Students then study supervised and unsupervised learning methods for regression, classification, and representation learning, including probabilistic models, decision trees, ensemble

methods, support vector machines, neural networks, and deep learning. Advanced topics include sequence models, transformers, reinforcement learning, and modern generative AI.

The course emphasizes understanding how learning algorithms work, selecting appropriate methods for different problems, evaluating model performance and generalization, and applying machine-learning techniques through programming and experimental analysis.

Prerequisites:

- Linear algebra knowledge
- A good working knowledge of Python or any program language
- Mathematics 1305
- Working knowledge of data structures, such as linked lists, trees, symbol tables, and dynamically
  allocated structures.
- No prior knowledge of machine learning is assumed

Course Learning Objectives

Upon successful completion of this course, students will be able to:

- Apply appropriate evaluation methods to identify overfitting, underfitting, and generalization issues and interpret experimental results.
- Apply probability, optimization, and other mathematical concepts that form the foundations of machine-learning algorithms.
- Describe the foundations of neural networks, deep learning, sequence models, transformers, generative AI, and reinforcement learning.
- Explain the limitations of linear models and how kernels, multilayer neural networks, and deep-learning methods address nonlinear problems.
- Explain the major types of machine-learning problems, including regression, classification, clustering, representation learning, and reinforcement learning.

- Read and critically evaluate machine-learning research and reproduce selected experimental results from published work.
- Train, evaluate, and compare classical machine-learning models, including nearest-neighbor methods, probabilistic models, decision trees, random forests, ensemble methods, and support vector machines.

Course Materials

Title: None
There is no primary text for this course. All required materials will be available online.

Grading Scale

Your semester grade for this course will be determined based on the following scale:

A = 90-100
B = 80-89
C = 70-79
D = 60-69
F = less than 60

A = excellent work; B = above average work; C = average work; D = below average; F = failing

Grading Criteria

Course Component	Percentage of Total Points
Lectures attendance, engagement and quizzes	10%
Assignments (4)	40%
Midterm Exam	20%
Final presentation	30%

Learning Activities

The course will combine interactive lectures, assignments, programming activities, examinations, guest lectures, and a research-based final project. The goal is to help students develop both conceptual understanding and practical experience with machine-learning methods, model evaluation, experimentation, and current research.

Interactive Lectures and In-Class Engagement

Lectures will include regular opportunities for active student participation and formative assessment. Students may respond to multiple-choice questions, solve examples, analyze model behavior, interpret experimental results, and participate in programming or algorithm-tracing activities.

Activities may include:

- Short MCQs assessing understanding of machine-learning concepts.
- Problem-solving exercises completed individually or in small groups.
- Programming and model-training activities.
- Interpretation of model predictions and evaluation metrics.
- Comparison of alternative algorithms or modeling approaches.
- Discussion of common implementation and experimental errors.

These activities are intended to connect theoretical concepts with practical machine-learning problems and provide immediate feedback before graded assignments and examinations.

Guest Lectures and Industry Engagement

Guest speakers from industry and academia may be invited to discuss current developments, research challenges, and practical applications of machine learning and artificial intelligence.

Students will gain exposure to current research, real-world applications, professional opportunities, and emerging challenges associated with machine-learning technologies.

Assignments

Students will complete four assignments during the semester. Assignments may include:

- Short technical reports.
- Problem-solving exercises.
- Programming assignments implementing or applying machine-learning algorithms.

Assignments will reinforce concepts discussed in class and provide experience with the complete machine-learning workflow.

Midterm Exam

A midterm examination will assess students' understanding of the major concepts covered during the first half of the course. The examination may include conceptual questions, problem-solving exercises, algorithm analysis, and interpretation of machine-learning methods and results.

Final Research Reproduction Project

The final project will introduce students to the process of reading, understanding, and experimentally evaluating current machine-learning research. Each student or team will work with an assigned or approved research paper and attempt to reproduce selected results reported in the paper.

The project will be completed in stages throughout the semester. Students will:

1. Read and understand the assigned research paper, including its motivation, proposed method, experimental setup, and main findings.
2. Identify the experiments or results to be reproduced.
3. Prepare the required datasets, software, models, and experimental environment.

4. Implement or execute the proposed method and document the reproduction process.
5. Collect and analyze their experimental results.
6. Compare their reproduced results with those reported in the original paper.
7. Discuss possible reasons for similarities or differences and provide their own technical observations and insights.

The goal is not necessarily to obtain identical numerical results. Students should demonstrate that they understand the machine-learning methodology, experimental design, evaluation process, and factors that may influence reproducibility.

Final Presentation

The final stage of the project will be an oral presentation during the final weeks of the semester. Students will be expected to:

- Explain the main idea, motivation, and significance of the assigned research paper.
- Clearly describe the machine-learning approach proposed in the paper.
- Explain the steps they followed to reproduce the selected experiments.
- Present their reproduced results and compare them with the original paper.
- Discuss differences, limitations, challenges, and observations from the reproduction process.
- Provide their own interpretation and insights based on the results.
- Answer questions and participate in a technical discussion about the paper, methodology, and experimental findings.

The final presentation will evaluate both the quality of the experimental work and the student's ability to understand, communicate, and critically discuss machine-learning research.

University Policies

Syllabus, Office Hours, and Regular and Substantive Contact

- https://ualr.edu/policy/facstaff/course-syllabus/

Students with Disabilities

- https://ualr.edu/policy/facstaff/course-syllabus/

If you are a student with a disability, or if you simply want to discuss resources that might help you learn more effectively, you can contact the Disability Resource Center at any time. You are welcome to drop in or call 501-916-3143 to make an appointment. Their staff will work with you to discuss accommodations and identify resources on campus or in the community that might be helpful for you. Accommodations are established through an interactive process. For more information, visit the DRC website or review the steps to request accommodations. We want you to know you are valued, welcome, and wanted at UA Little Rock.

Non-Discrimination Policy

- https://ualr.edu/policy/home/admin/non-discrimination/

Title IX and Harassment

- https://ualr.edu/titleix/titleix/title-ix-the-basics/

Inclement Weather Policy

- https://ualr.edu/policy/admin/weather/

Academic Integrity

- https://ualr.edu/deanofstudents/academic-integrity/

Disclosure of Instances of Sexual Misconduct

- https://ualr.edu/policy/home/facstaff/title-ix/

Withdrawal

- https://ualr.edu/policy/home/student/withdrawal-from-ualr/

Add/Drop

- https://ualr.edu/records/2511-2/

Incomplete

- https://ualr.edu/records/grades/

Grade Policies

- https://ualr.edu/policy/home/facstaff/grades-and-grading-systems/

Course Policies

Course Attendance, Engagement and Expectations

Attendance matters!

This course emphasizes active learning through discussion, short exercises, quizzes, problem solving, programming activities, and the reproduction of established research papers. Students are expected to attend class, participate actively, and complete assigned work on time.

Attendance and participation are important because many course concepts will be reinforced through in-class examples, discussions, and problem-solving activities. Students who cannot attend should contact the instructor and review the posted course materials to remain up to date.

Use of Generative AI

The use of generative AI tools, including large language models (LLMs) such as ChatGPT and similar tools, is generally permitted as a learning aid unless

otherwise stated for a specific assignment or activity.

Generative AI should be used as a tool to support learning, not as a substitute for learning. Students remain fully responsible for everything they submit, including its accuracy, correctness, originality, references, code, and interpretation of results.

Students should therefore:

- Review, verify, and understand any AI-generated content before using it.
- Never submit material that they cannot explain or defend.
- Verify generated code, calculations, references, and technical claims rather than assuming they are correct.
- Never use generative AI to fabricate experimental results, references, data, or other evidence.
- Follow any additional AI-use instructions provided for individual assignments or activities.

The instructor may ask students to explain their submitted work, code, methodology, or results during class, office hours, project presentations, or other discussions. Students should be prepared to demonstrate that they understand the work they submitted and the decisions they made.

Generative AI tools may not be used during the midterm examination or other individual assessments where their use is explicitly prohibited.

Academic Responsibility

Collaboration and the use of learning resources are encouraged when appropriate, but submitted work should reflect the student's own understanding and effort. Students are responsible for following assignment-specific instructions and the University's academic integrity requirements.

Syllabus and Course Adjustments

This syllabus serves as a roadmap for the course. Topics, activities, assignment dates, or other course elements may be adjusted based on class progress, student needs, or developments in the field of machine learning and artificial intelligence. Any significant changes will be communicated clearly through class announcements, email, or Blackboard.

Helpful University Resources

Blackboard Student Support

- https://ualr.edu/blackboard/contact/

Bookstore

- https://ualr.bncollege.com/

Care Team

- https://ualr.edu/studentsuccess/care-team/

Career Center

- https://ualr.edu/careers/

Child Care Connections

- https://ualr.edu/studentsuccess/child-care-connections/

Disabilities Resource Center

- https://ualr.edu/disability/

Communication Skill Center (CSC)

- https://ualr.edu/appliedcomm/communication-skill-center-csc/

Counseling

- https://ualr.edu/counseling/

Handshake

- https://ualr.joinhandshake.com/login

Health Services

- https://ualr.edu/health/

Information Technology Services (ITS)

- https://ualr.edu/itservices/welcome/

Mathematics Assistance Center

- https://ualr.edu/mathematics/mathematics-assistance-center/

Military Student Success Center

- https://ualr.edu/military/

Office of Student Retention Initiatives (SRI)

- https://ualr.edu/studentsuccess/academic-resources/coaching-and-student-support/

Student Support Services (SSS)

- https://ualr.edu/studentsupport/

Trojan Tutoring (i.e. introductory-level courses only) via University Learning Commons

- https://ualr.edu/studentsuccess/academic-resources/

UA Little Rock Trojan Food Pantry

- https://ualr.edu/foodpantry/

University Writing Center (UWC)

- https://ualr.edu/writingcenter/

Ottenheimer Library

- https://ualr.edu/library/

Tentative Course Schedule

Subject to change at the discretion of instructor. The schedule is tentative and may be adjusted based on class progress and instructional needs.

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Topic	**Week**
Introduction to Artificial Intelligence and Machine Learning	1
Search	2
Knowledge and Logical Reasoning	3
Uncertainty and Probabilistic Reasoning	4
Optimization and Constraint Satisfaction	5
Machine Learning Foundations and Classical Models	6
Midterm Examination	7
Linear and Nonlinear Machine Learning	8
Neural Networks and Deep Learning	9
Language and Modern AI	10
Reinforcement Learning	11
Final Project Presentations	12 - 15